

EE4725 - Quasi Optical Systems - Final Projects

Project 1.1, 1.2

The Green Bank Telescope is made of a parabolic antenna with a 100m diameter and F/D ratio of 0.43. The reflector is fed with a dipole antenna operating at f_0 , with

Project 1.1	Project 1.2
$f_0 = 1 \text{ GHz}$	$f_0 = 3 \text{ GHz}$

Determine:

1. The subtended angle of the reflector
2. The dimensions of the dipole that will maximize the aperture efficiency of the telescope
3. The far field of the parabolic antenna and its directivity for those dimensions
4. The taper, spill over and aperture efficiencies for those dimensions
5. How would be the far field of an ideal telescope feed that would lead to a telescope aperture efficiency of 100%? How would the reflector far field be for such case?



Justify your answers with figures and try to give a physical insight into the problem.

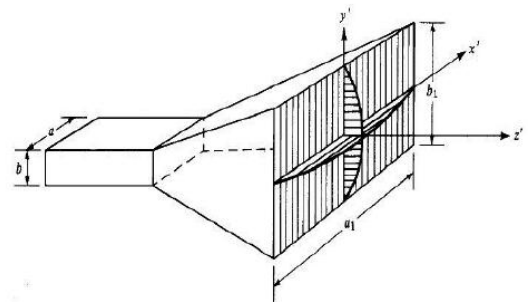
Project 2.1, 2.2

One of the most widely used reflector feeds is a pyramidal horn antenna. It is made by flaring a rectangular waveguide in the two main planes as shown in the figure.

The electric and magnetic tangent fields at the aperture of a pyramidal horn antenna can be approximated by the following expressions:

$$E_y(x', y', z') = E_0 \cos\left(\frac{\pi}{a_1} x'\right) e^{-jk_0\left(\frac{x'^2}{\rho_1} + \frac{y'^2}{\rho_2}\right)}$$

$$H_x(x', y', z') = -\frac{E_0}{\eta} \cos\left(\frac{\pi}{a_1} x'\right) e^{-jk_0\left(\frac{x'^2}{\rho_1} + \frac{y'^2}{\rho_2}\right)}$$



Consider a frequency of f_0 and the horn parameters to be a_1 , b_1 , and $\rho_1 = \rho_2$. This horn is used to feed a reflector antenna of 5m in diameter.

Project 2.1	Project 2.2
$f_0 = 10 \text{ GHz}$	$f_0 = 20 \text{ GHz}$
$a_1 = 124 \text{ mm}, b_1 = 92 \text{ mm},$	$a_1 = 62 \text{ mm}, b_1 = 46 \text{ mm},$
$\rho_1 = \rho_2 = 255 \text{ mm}$	$\rho_1 = \rho_2 = 125 \text{ mm}$

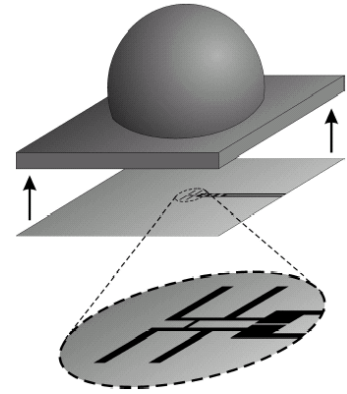
1. Find the far fields radiated by the horn alone together with its broadside directivity. Where is the phase center of the horn?
2. Evaluate the power received from a broadside plane wave incident on the reflector antenna as a function of the focal distance and the z-location of the horn. What is the optimum focal distance and z-location?
3. Evaluate the power received now as a function of the plane wave incident angle for the optimum case. Give the values of directivity, spill over and aperture efficiency

Justify your answers with figures and try to give a physical insight into the problem.

Project 3.1, 3.2

It is common to use an elliptical silicon lens antenna to be integrated with a high frequency detector. The detector, which operates at f_0 , it is placed on silicon ($\epsilon_r = 11.9$). To couple the radiation into the detector an antenna consisting of two slots is used.

Project 3.1	Project 3.2
$f_0 = 350$ GHz	$f_0 = 200$ GHz



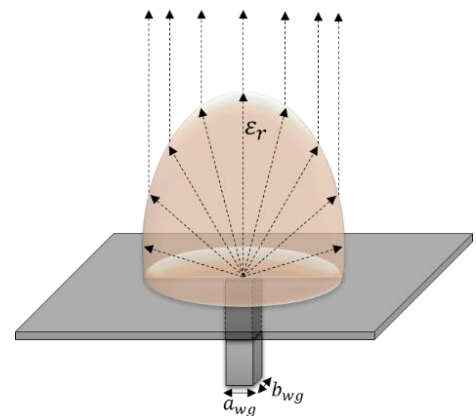
1. Give the length and separation of the slots needed to generate a field radiated inside an infinite silicon medium that could be used to illuminate efficiently the lens.
2. Design the lens dimensions such that its far field is characterized by a directivity of 30 dBi.
3. What is the side lobe level, directivity and reflection efficiency of lens antenna?
4. Assuming we aimed at the same directivity of the lens antenna, how would the two slot antenna and lens dimensions change if the dielectric material was $\epsilon_r = 4$? What about reflection efficiency?

Justify your answers with figures and try to give a physical insight into the problem.

Project 4.1, 4.2

For automotive radar applications at f_0 , we need to design an elliptical lens antenna with a directivity of 30 dBi fed by a rectangular waveguide as shown in the figure. The waveguide can be modelled considering the field in the aperture is approximated by the TE₁₀ mode.

Project 4.1	Project 4.2
$f_0 = 70$ GHz	$f_0 = 100$ GHz



1. Make a study of the optimal dimensions of the waveguide and the lens to reach the specifications for the following different lens materials: silicon ($\epsilon_r = 11.9$), quartz ($\epsilon_r = 4$) and plastic ($\epsilon_r = 2$)
2. How does the change physical dimensions with the lens dielectric material?
3. How does the critical angle change with the lens dielectric material? And the directivity inside the dielectric and the reflection efficiency of the optimized waveguide feed?
4. How do the radiated patterns of the different lens antennas compare to each other?

Justify your answers with figures and try to give a physical insight into the problem.

Project 5.1, 5.2

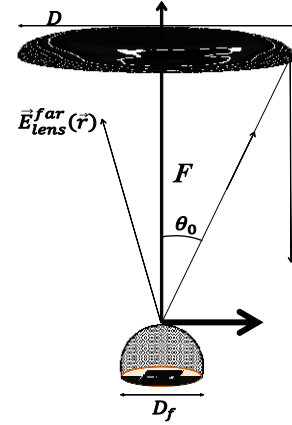
For astronomical applications, a ($\epsilon_r = 11.9$) silicon lens antenna is used to illuminate a parabolic reflector of 50cm diameter. The lens feed is a planar antenna operating at f_0 that can be modelled with the following primary pattern:

$$E_{\text{feed}}^{\text{lens}}(r, \theta, \phi) = \cos^n \theta (\cos \phi \hat{\theta} - \sin \phi \hat{\phi}) \frac{e^{-jk_d r}}{r}$$

Project 5.1	Project 5.2
$f_0 = 600\text{GHz}$	$f_0 = 400\text{GHz}$

1. Choose the feed n parameter and the lens dimensions that maximizes the aperture efficiency of the parabolic antenna.
2. Provide the lens and reflector far-field directivity patterns and efficiencies (spill over, taper, reflection) for the chosen parameters.
3. Assuming that the lens feed pattern is frequency independent, how would the gain of the overall antenna system change in the frequency range of $0.5f_0$ - $1.5f_0$?

Justify your answers with figures and try to give a physical insight into the problem.



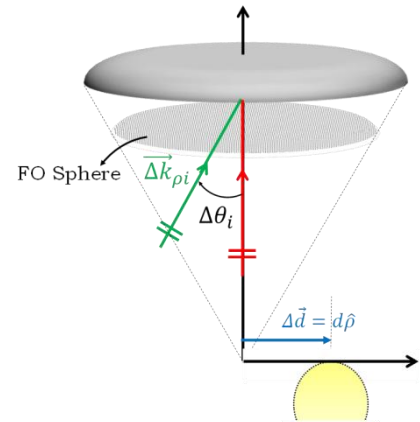
Project 6

For sensing applications, we want to detect the EM radiation at f_0 incident into a parabolic antenna (diameter of $D=15\text{cm}$ and focal distance of $F=40\text{cm}$) from a specific angular direction ($\theta_i = 5^\circ$, $\phi_i = 0^\circ$) and TM polarization. For detecting this radiation, we can use an integrated detector into a silicon lens antenna ($\epsilon_r = 11.9$ and a major axis of $a = 7.5\text{mm}$). Consider the antenna at the lens focus to have a radiation pattern of

$$E_{\text{feed}}^{\text{lens}}(r, \theta, \phi) = \cos^n \theta (\cos \phi \hat{\theta} - \sin \phi \hat{\phi}) \frac{e^{-jk_d r}}{r}$$

with $n = 2$

Project 6.1	Project 6.2
$f_0 = 150\text{GHz}$	$f_0 = 350\text{GHz}$



Evaluate the far field radiated by the lens, directivity and its reflection efficiency

1. Evaluate the optimum location of the lens antenna in the focal plane, $\Delta \vec{d}$, of the reflector that will maximize the power coming from the specific plane wave direction of $\theta_i = 5^\circ$, $\phi_i = 0^\circ$. What does this location correspond to?
2. Evaluate the power received now, for that specific lens position, as a function of the plane wave incident angle. Give the values of directivity, spill over and aperture efficiency of the considered quasi-optical system.
3. Is it possible to improve the received power by changing the reflector or lens dimensions?

Justify your answers with figures and try to give a physical insight into the problem.

Project 7.1, 7.2

For space telecommunication links operating at f_0 , a focal plane array of horn antennas coupled to a parabolic reflector of 1m diameter is used to generate multi-beams that cover part of the Earth. Assume that each horn can be modelled with the following aperture fields in a reference system centered at the horn:

$$E_y(x', y', z') = E_0 \cos\left(\frac{\pi}{a} x'\right) \text{rect}(y', b)$$

$$H_x(x', y', z') = -\frac{E_0}{\eta} \cos\left(\frac{\pi}{a} x'\right) \text{rect}(y', b)$$

Let us consider an array of horns where the spacing is given by $d_x = d_y = a = b = 3\lambda_0$.



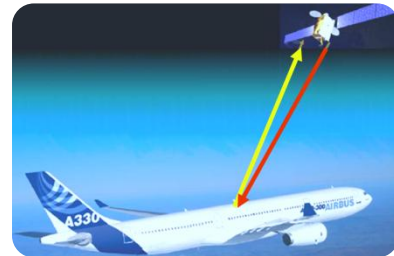
Project 7.1	Project 7.2
$f_0 = 30\text{GHz}$	$f_0 = 20\text{GHz}$

1. Choose the reflector focal distance that leads to the highest aperture efficiency for the central horn.
2. Evaluate the reflector patterns, in such optimum focal distance, for the central horn and the neighbor horns of the focal plane array. What is the beam overlap level between the different patterns?
3. What would be the focal distance required to have a beam overlap of -3dB? What would be the aperture efficiency in this case?
4. How would you design a quasi-optical system that leads to high aperture efficiency and a -3dB beam overlap?

Justify your answers with figures and try to give a physical insight into the problem.

Project 8.1, 8.2

Consider the design of an antenna array to be integrated into an aircraft for satellite communication (Satcom). The array consists of dipoles in the presence of a backing reflector and must operate in the Satcom transmit frequency band.



Project 8.1	Project 8.2
Ku-band: 14-14.5 GHz	Ka-band: 27.5-31 GHz

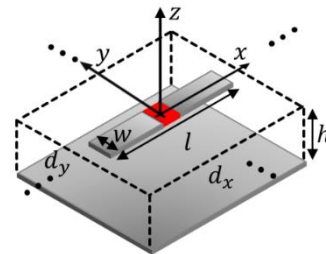
1. Find the appropriate array parameters to achieve active reflection coefficient

$$\Gamma_{\text{act}}(\text{dB}) = 20 \log_{10} \left| \frac{Z_{\text{in,active}} - Z_L}{Z_{\text{in,active}} + Z_L} \right| < -10\text{dB}$$

within the operational frequency band. The design parameters you can vary are the following: array element dimensions (h, w, l), array periods (d_x, d_y), characteristic impedance of the feed (Z_L).

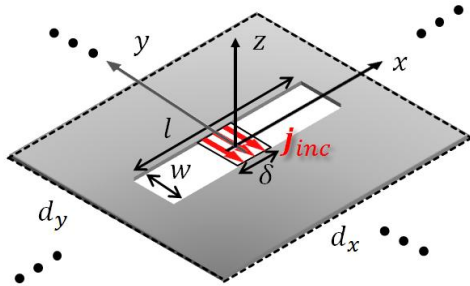
2. By observing the variation of Γ_{act} with scanning angle, what is the maximum scan angle that can be achieved by the array to avoid grating lobes and scan blindness?
3. How many elements are needed in the array to achieve half power beamwidth of 2 degrees?

Justify your answers with figures and motivate your choices, trying to give a physical insight into the problem.

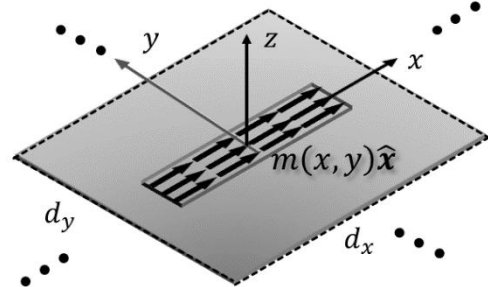


Project 9

Consider an infinite, doubly periodic array of slots in free space. Assume that the slots are excited with a surface current density $\mathbf{j}_{inc} = I_0 \frac{\text{rect}_{\text{gap}}(x,y)}{\delta} \hat{\mathbf{y}}$, distributed on a feed of dimension $\delta \ll \lambda$. The slot has width $w \ll \lambda$ and length l , and they are periodic with periods d_x and d_y along x and y respectively.



Original problem



Equivalent problem

By applying the equivalence theorem, the integral equation to be solved to find the unknown magnetic current distribution is

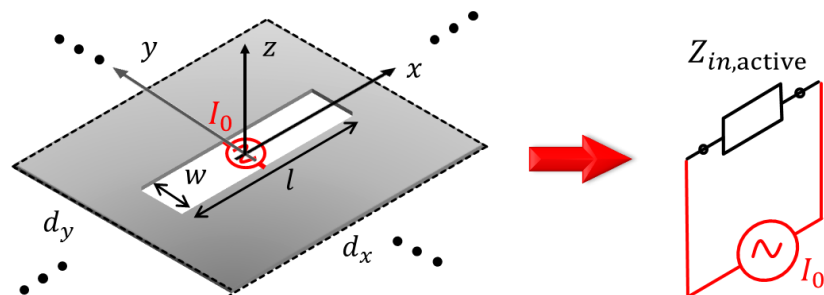
$$-4g_{xx,FS}^{hm}(x,y) * m(x,y) = I_0 \frac{\text{rect}_{\text{gap}}(x,y)}{\delta}$$

The magnetic current distribution on the slot can be assumed to be sinusoidal:

$$m(x,y)\hat{\mathbf{x}} = V_0 b(x,y)\hat{\mathbf{x}} = V_0 \frac{\sin\left(k_0\left(\frac{l}{2} - |x|\right)\right)}{\sin\left(k_0\frac{l}{2}\right)} \frac{\text{rect}_{\left[\frac{-w}{2}, \frac{w}{2}\right]}(y)}{w} \hat{\mathbf{x}}$$

where V_0 is the unknown amplitude of the basis function.

1. Derive an expression for the active input admittance of the slot in free space, showing the steps followed in the derivation.



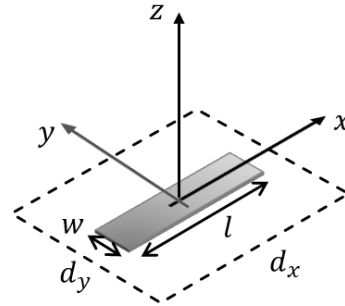
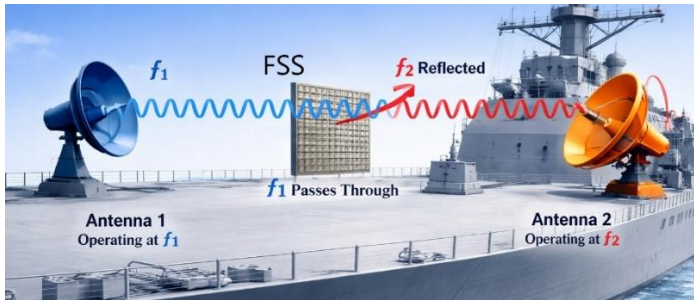
2. Plot the variation of real and imaginary part of the impedance with the frequency (in the range 5-15 GHz) for $d_x = d_y = 15\text{mm}$, $w = 3\text{mm}$, $l = 14\text{mm}$. How does the active impedance of a slot in infinite array compare with the one of dipole with the same dimensions?
3. Plot the fundamental mode and higher order modes of the active impedance, for scanning to $\phi = 0^\circ, \theta = 0^\circ$ and to $\phi = 0^\circ, \theta = 60^\circ$ and comment on the plots. Are the quantities real or imaginary? What happens when a grating lobe appears and why?

Project 10.1 and 10.2

On a military ship, there are two antenna systems operating at the frequencies f_1 and f_2 , respectively. You want to design an FSS radome to be placed in front of the first antenna (operating at f_1) to avoid interference with the second antenna (operating at f_2).



Project 10.1	Project 10.2
$f_1 = 6 \text{ GHz}$	$f_1 = 8 \text{ GHz}$
$f_2 = 12 \text{ GHz}$	$f_2 = 3 \text{ GHz}$



1. If the FSS consists of an array of metal strips, describe how you would select the array element dimensions (w, l) and the array periods (d_x, d_y) to design the FSS.
2. How do the reflection and transmission coefficient of the FSS change with the angle of incidence, for TE and TM cases?
3. If the FSS is made of 30×30 elements, how do the patterns change with the angle of incidence?

Justify your answers with figures and motivate your choices, trying to give a physical insight into the problem.

Project 11

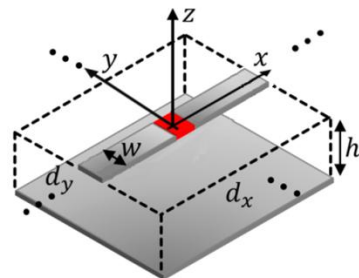
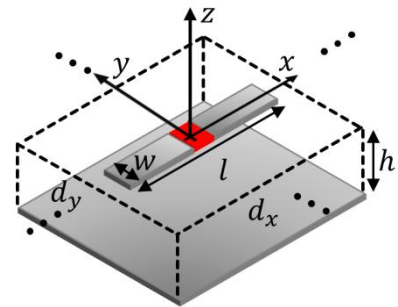
Consider an infinite array of dipoles in the presence of a backing reflector. The geometrical parameters of the unit cell are shown in the figure.

1. Study the bandwidth of the resonant dipoles, with sinusoidal current distribution, by plotting the active reflection coefficient

$$\Gamma_{\text{act}}(\text{dB}) = 20 \log_{10} \left| \frac{Z_{\text{in,active}} - Z_L}{Z_{\text{in,active}} + Z_L} \right|$$

as a function of the frequency and for different scanning angles. On what frequency range and what scan range can you obtain $\Gamma_{\text{act}}(\text{dB}) < -10\text{dB}$?

A connected dipole array consists of dipoles with electrical connections between elements, like in figure. The purpose is to obtain a current distribution that is more constant over the dipole rather than sinusoidal, to improve bandwidth.



Assuming that the current distribution is not anymore sinusoidal but constant:

$$b(x, y) = \text{rect}\left[\frac{-l}{2}, \frac{l}{2}\right](x) \frac{\text{rect}\left[\frac{-w}{2}, \frac{w}{2}\right](y)}{w}$$

2. Compute and plot the active input impedance of the connected dipole array, as a function of the frequency and the scan angle.
3. Compare the two radiating elements in terms of bandwidth and scan range, by plotting the active input impedance and the active reflection coefficient for the two cases.